E-companion for ‘Proficiency Boosts Efficiency: Extended Johnson’s Rules for PFSP with CLFEs’

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EC.1 Summary of Notations and Abbreviations

For any integer $K \in \mathbb{Z}_+$, we define $[K] \triangleq \{1, \cdots, K\}$, which is the set of positive running indices to K. The list of notations used in this paper is shown in Table EC.1.

Table EC.1 Notations List

Sets	
$\mathcal{J}$	Job set, $j \in \mathcal{J} = [n]$, where n denotes the number of jobs
$\mathcal{M}$	Machine set, $m \in \mathcal{M} = [M]$, where M denotes the number of machines
$\mathcal{C}$	Category set, $c \in \mathcal{C} = [G]$, where G denotes the number of categories
$\mathcal{G}$	Batch set, where each $c \in \mathcal{G} = [G]$ denotes a batch of category c
$\mathcal{J}_1, \mathcal{J}_2$	Job subsets, $\mathcal{J}_1 = \left\{j \in \mathcal{J} \mid t_{j,1} \leq t_{j,2}\right\} \subseteq \mathcal{J}$ and $\mathcal{J}_2 = \left\{j \in \mathcal{J} \mid t_{j,1} > t_{j,2}\right\} \subseteq \mathcal{J}$
$\mathcal{G}_1, \mathcal{G}_2$	Batch subsets, $\mathcal{G}_1 = \{c \in \mathcal{G} \mid P_{c,1} \leq P_{c,2}\} \subseteq \mathcal{G}$ and $\mathcal{G}_2 = \{c \in \mathcal{G} \mid P_{c,1} > P_{c,2}\} \subseteq \mathcal{G}$
Parameters for Categories and Jobs	
n_c	The number of jobs belonging to category c
$d_{j,c}$	Binary indicator for whether job j belongs to category c
$t_{j,m}$	The normal processing time of the job j on machine m
$P_{c,m}$	The normal processing time of jobs in category c on machine m
$C^{\pi}_{k,m}$	The completion time of the k-th job on machine m in a schedule π
$C^{\pi}_{\max}$	The makespan of the schedule π
Parameters for Learning and Forgetting Effects	
α	Learning rate of the learning effect, $\alpha \in (0, 1]$
δ	Forgetting factor of the forgetting effect, $\delta \geq 0$
β_k	Effective learning position of the category of the k-th job in a schedule
Function	
$f(\beta)$	The general learning-forgetting function, $f : [1, \infty) \to (0, 1]$
Notation Simplifications for Batches and Bacth-wise Schedules	
$P^{(i)}_{c,m}$	$\triangleq P_{c,m} \cdot f(i), \quad \forall i \geq 1, c \in \mathcal{C}, m \in \mathcal{M}$
$P'_{c,m}$	$\triangleq P^{(n_c)}_{c,m}, \quad \forall c \in \mathcal{C}, m \in \mathcal{M}$
$\bar{P}_{c,m}$	$\triangleq \sum_{i\in[n_c]} P^{(i)}_{c,m}, \quad c \in \mathcal{C}, m \in \mathcal{M}$
$\bar{C}_{g,m}$	The completion time of the g-th batch in a batch-wise schedule
$\bar{C}^{\pi}_{g,m}$	The completion time of the g-th batch in a batch-wise schedule π
Decision Variables	
$x_{j,k}$	Binary variable, $x_{j,k} = 1$ means the job j is the k-th processed job, $\forall j \in \mathcal{J}, k \in [n]$
$u_{k,c}$	Continuous variable, the effective learning position of category c at the k-th job in a schedule, $\forall k \in [n], c \in \mathcal{C}$
$p_{k,m}$	Continuous variable, the actual processing time of the k-th job on machine m in a schedule, $\forall k \in [n], m \in \mathcal{M}$
$C_{k,m}$	Continuous variable, the completion time of the k-th job on machine m in a schedule, $\forall k \in [n], m \in \mathcal{M}$
$C_{\max}$	Continuous variable, the makespan of a schedule

The abbreviations and their full forms used in this paper are listed in Table EC.2.

Table EC.2 List of Abbreviations and Full Terms

Abbreviations	**Full Terms**
PT	Processing Time
PFSP	Permutation Flow-shop Scheduling Problem
LFEs	Learning and Forgetting Effects
CLFEs	Category-dependent Learning and Forgetting Effects
$F2/\text{Prmu}/C_{\max}$	Two-machine permutation flow shop scheduling problem that minimizes the makespan
$F2/\text{Prmu, CLFEs}/C_{\max}$	the $F2/\text{Prmu}/C_{\max}$ problem with CLFEs
$Fm/\text{Prmu, CLFEs}/C_{\max}$	Multi-machine permutation flow shop scheduling problem with CLFEs that minimizes the makespan
$F2/\text{Prmu,CLSFEs}/C_{\max}$	$F2/\text{Prmu, CLFEs}/C_{\max}$ with learning and strong forgetting effects
$F2/\text{Prmu,CLE}/C_{\max}$	$F2/\text{Prmu, CLFEs}/C_{\max}$ without the forgetting effect
BJ Rule	Batch-wise Johnson's Rule
PJ Rule	Partition-based Johnson's Rule
MBJ Algorithm	Multi-machine Batch-wise Johnson's Algorithm

EC.2 Makespan of Given Schedules

For a given schedule for the $Fm/\text{Prmu, CLFEs}/C_{\max}$ problem, the completion time of each job is calculated using the recursive formula stated in Lemma EC.1.

Lemma EC.1. (Reza Hejazi and Saghafian 2005) *In the* $Fm/\text{Prmu, CLFEs}/C_{\max}$ *problem, the completion time of the* k*-th job* $(k \in [n])$ *processed on the machine* m $(m \in \mathcal{M})$ *in a schedule is given by*

$$\begin{aligned} C_{k,m} &= \max\{C_{k,m-1}, C_{k-1,m}\} + p_{k,m} \\ &= \max_{1\le a_1\le\cdots\le a_m=k}\left\{\sum_{i=1}^{a_1} p_{i,1} + \cdots + \sum_{i=a_{m-1}}^{a_m} p_{i,m}\right\}, \end{aligned}$$

where

$$C_{k,1} = \sum_{i=1}^{k} p_{i,1}, \quad C_{0,m} = 0, \quad \forall m \in \mathcal{M}.$$

In a batch-wise schedule of the $Fm/\text{Prmu, CLFEs}/C_{\max}$ problem, Lemma EC.2 provides formulas for the completion times of each batch on the machines. Let $\bar{C}_{g,m}$ be the completion time of the g-th batch on the machine m in a batch-wise schedule, and $P_{c,m}^{(i)}$ be the actual PT of a job of category c at the effective learning position i on machine m, i.e., $P_{c,m}^{(i)} \triangleq P_{c,m} \cdot f(i)$ for any $i \ge 1, c \in \mathcal{C}, m \in \mathcal{M}$. Figure EC.1 illustrates all possible scenarios for these completion times when $m = 2$.

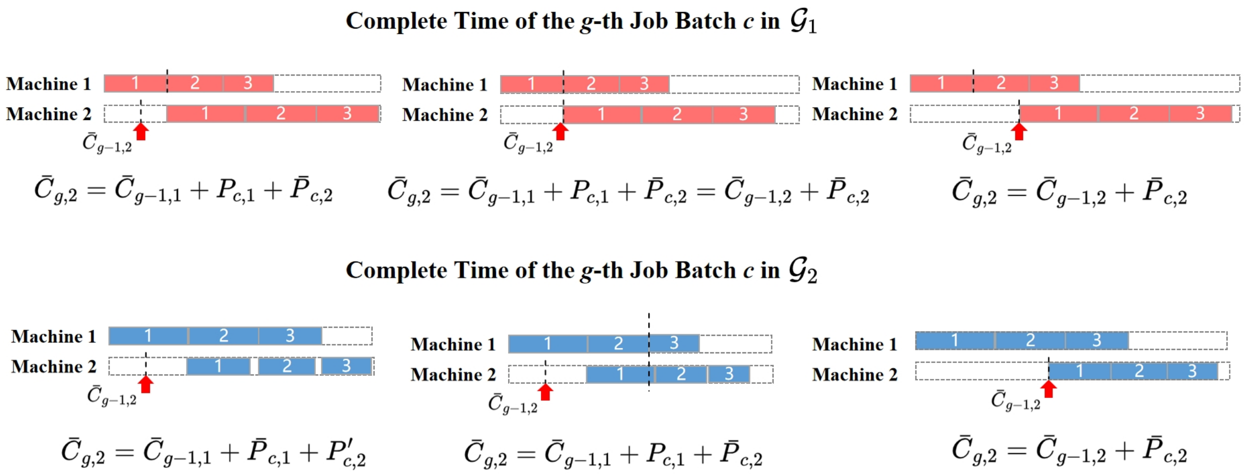

Figure EC.1 An Intuitive Illustration for Lemma EC.2 when $m = 2$

LEMMA EC.2. *For the g-th processed batch $c\,(c \in \mathcal{G})$ with r jobs in a batch-wise schedule of the Fm/*Prmu, CLFEs/$C_{\max}$ *problem $(1 \le g \le G, 1 \le r \le n_c)$, its completion time on machine m $(m \in \mathcal{M})$, $\bar{C}_{g,m}$, is given by:*

- *if* $m = 1$: $\bar{C}_{g,1} = \bar{C}_{g-1,1} + \bar{P}_{c,1}$;
- *if* $m = 2$:
  — *if* $P_{c,1} \le P_{c,2}$, *then*

$$\bar{C}_{g,2} = \max\left\{\bar{C}_{g-1,1} + P_{c,1} + \bar{P}_{c,2}, \bar{C}_{g-1,2} + \bar{P}_{c,2}\right\};$$

  — *if* $P_{c,1} > P_{c,2}$, *then*

$$\bar{C}_{g,2} = \max\left\{\begin{array}{l} \bar{C}_{g-1,1} + P_{c,1} + \bar{P}_{c,2}, \\ \bar{C}_{g-1,1} + \bar{P}_{c,1} + P'_{c,2}, \\ \bar{C}_{g-1,2} + \bar{P}_{c,2} \end{array}\right\};$$

  where

$$\bar{C}_{g,1} = \bar{C}_{g-1,1} + \bar{P}_{c,1};$$

- *if* $m \ge 3$:

$$\bar{C}_{g,m} = \max\left\{\begin{array}{l} S_{g-1}, \\ \bar{C}_{g-1,m-1} + P_{c,m-1} + \bar{P}_{c,m}, \\ \bar{C}_{g-1,m-1} + \bar{P}_{c,m-1} + P'_{c,m}, \\ \bar{C}_{g-1,m} + \bar{P}_{c,m} \end{array}\right\},$$

  where

$$S_{g-1} = \max_{\substack{\forall l \in [m-2], \\ u_1, u_2, \ldots, u_{m-l} \in \{1, r\}, \\ u_1 \le u_2 \le \cdots \le u_{m-l}}} \left\{\bar{C}_{g-1,l} + \sum_{i=1}^{u_1} P^{(i)}_{c,l} + \sum_{i=u_1}^{u_2} P^{(i)}_{c,l+1} + \cdots + \sum_{i=u_{m-l}}^{r} P^{(i)}_{c,m}\right\};$$

where $\bar{C}_{0,m} = 0$, $\forall m \in \mathcal{M}$.

Proof of Lemma EC.2. Here, we provide the detailed proof for the case where $m = 2$. The proofs for other cases follow a similar, albeit more complex, logic and are omitted for brevity. For the g-th processed batch with r jobs in a batch-wise schedule of $F2/\text{Prmu, CLFEs}/C_{\max}$ (where $g \geq 1$ and $r \geq 1$), according to Lemma EC.1, its completion time $\bar{C}_{g,2}$ is given by

$$\bar{C}_{g,2} = \max\left\{ \begin{array}{l} S_{1,g,u} \triangleq \bar{C}_{g-1,1} + \sum_{i=1}^{u} P_{c,1}^{(i)} + \sum_{i=u}^{r} P_{c,2}^{(i)}, \quad \forall u \in [r], \\ S_{2,g} \triangleq \bar{C}_{g-1,2} + \bar{P}_{c,2} \end{array} \right\}, \quad \forall g \in [G], \tag{EC.1}$$

where

$$\bar{C}_{g,1} = \bar{C}_{g-1,1} + \bar{P}_{c,1}, \quad \forall g \in [G],$$

$$\bar{C}_{0,m} = 0, \quad \forall m \in \mathcal{M}.$$

The start time of g-th batch on the second machine $R_{g,2}$ in a batch-wise schedule for the $F2/\text{Prmu, CLFEs}/C_{\max}$ is given by Eq. (EC.2):

$$R_{g,2} = \max\left\{ \bar{C}_{g-1,1} + P_{c,1}^{(1)}, \bar{C}_{g-1,2} \right\}, \forall g \in [G]. \tag{EC.2}$$

According to Lemma EC.1, the PT of g-th batch on the second machine is $\bar{P}_{c,2}$ when $r = 1$. Otherwise, it is given by

$$\max\left\{ \sum_{i=1}^{r-1} P_{c,2}^{(i)}, \sum_{i=2}^{r} P_{c,1}^{(i)} \right\} + P_{c,2}^{(r)} = \max\left\{ \bar{P}_{c,2}, \bar{P}_{c,1} - P_{c,1}^{(1)} + P_{c,2}^{(r)} \right\}. \tag{EC.3}$$

Therefore, the completion time $\bar{C}_{g,2}$ of the g-th batch can be simplified, as shown in the following two cases.

- **Case 1:** $P_{c,1} \leq P_{c,2}$**.**

  According to the learning effect, we have

  $$P_{c,2}^{(i)} \geq P_{c,1}^{(i)} \geq P_{c,1}^{(i+1)}, \quad \forall i \geq 1.$$

  Thus, for any $r > 1$, we have

  $$\sum_{i=1}^{r-1} P_{c,2}^{(i)} \geq \sum_{i=2}^{r} P_{c,1}^{(i)}.$$

  Therefore, in this case, regardless of the value of r, the total PT of the batch on the second machine is always $\bar{P}_{g,2}$. Hence, the completion time of the g-th batch on the second machine is:

  $$\bar{C}_{g,2} = R_{g,2} + \bar{P}_{c,2} = \max\left\{ \bar{C}_{g-1,1} + P_{c,1} + \bar{P}_{c,2}, \bar{C}_{g-1,2} + \bar{P}_{c,2} \right\}. \tag{EC.4}$$

- **Case 2:** $P_{c,1} > P_{c,2}$**.**

  Based on Eqs. (EC.2) and (EC.3), the completion time of this batch on the second machine is

  $$\begin{aligned} \bar{C}_{g,2} &= \max\left\{ \bar{C}_{g-1,1} + P_{c,1}^{(1)}, \bar{C}_{g-1,2} \right\} + \max\left\{ \bar{P}_{c,2}, \bar{P}_{c,1} - P_{c,1}^{(1)} + P_{c,2}^{(r)} \right\} \\ &= \max\left\{ \begin{array}{l} \bar{C}_{g-1,1} + P_{c,1}^{(1)} + \bar{P}_{c,2}, \\ \bar{C}_{g-1,1} + \bar{P}_{c,1} + P_{c,2}^{(r)}, \\ \bar{C}_{g-1,2} + \bar{P}_{c,2}, \\ \bar{C}_{g-1,2} + \bar{P}_{c,1} - P_{c,1}^{(1)} + P_{c,2}^{(r)} \end{array} \right\} = \max\left\{ \begin{array}{l} \bar{C}_{g-1,1} + P_{c,1}^{(1)} + \bar{P}_{c,2}, \\ \bar{C}_{g-1,1} + \bar{P}_{c,1} + P_{c,2}^{(r)}, \\ \bar{C}_{g-1,2} + \bar{P}_{c,2} \end{array} \right\}. \end{aligned} \tag{EC.5}$$

In the last step of Eq. (EC.5), the fourth term in the expanded maximum function, $\bar{C}_{g,2} = \bar{C}_{g-1,2} + \bar{P}_{c,1} - P_{c,1}^{(1)} + P_{c,2}^{(r)}$, is dominated and can be removed. This term only occurs if $\bar{C}_{g-1,1} + P_{c,1}^{(1)} \geq \bar{C}_{g-1,2}$. However, under this case, the inequality $\bar{C}_{g-1,1} + P_{c,1}^{(1)} + \bar{P}_{c,1} - P_{c,1}^{(1)} + P_{c,2}^{(r)} \geq \bar{C}_{g-1,2} + \bar{P}_{c,1} - P_{c,1}^{(1)} + P_{c,2}^{(r)}$ holds, which implies that the term $\bar{C}_{g-1,1} + \bar{P}_{c,1} + P_{c,2}^{(r)}$ is always be greater than or equal to $\bar{C}_{g-1,2} + \bar{P}_{c,1} - P_{c,1}^{(1)} + P_{c,2}^{(r)}$.

From Eqs. (EC.4) and (EC.5), the case $m = 2$ in Lemma EC.2 is proved. ■

EC.3 Proofs for Main Results

Let $P_{c,m}^{(i)}$ denote the actual PT of a job of category c at the effective learning position i on machine m, i.e.,

$$P_{c,m}^{(i)} \triangleq P_{c,m} \cdot f(i), \forall c \in \mathcal{C}, m \in \mathcal{M}, i \geq 1.$$

Let $C_{k,m}^{\pi}$ denote the completion time of the k-th job on machine m in a schedule π, and $C_{\max}^{\pi}$ denote the makespan of the schedule π.

EC.3.1 Proof of Theorem 1.

To prove Theorem 1, we provide some preliminary analyses through the following Lemma EC.3 and Lemma EC.4.

For any batch-wise schedule π_1 of the $F2/\text{Prmu, CLFEs}/C_{\max}$ problem with G job categories, let the category of the last two batches be a and b, containing r and t jobs respectively ($r \geq 1, t \geq 1$). By swapping the positions of batches a and b, we obtain a new schedule π_2. The following Lemma EC.3 holds.

Lemma EC.3. *For the batch-wise schedules π_1 and π_2 defined above*:

(1) if $a \in \mathcal{G}_1$ and $b \in \mathcal{G}_2$, then $C_{\max}^{\pi_1} \leq C_{\max}^{\pi_2}$;

(2) if $a \in \mathcal{G}_2$ and $b \in \mathcal{G}_1$, then $C_{\max}^{\pi_1} \geq C_{\max}^{\pi_2}$;

(3) if $a \in \mathcal{G}_1$ and $b \in \mathcal{G}_1$:

- *when $P_{a,1} \leq P_{b,1}$ holds, then $C_{\max}^{\pi_1} \leq C_{\max}^{\pi_2}$;*
- *when $P_{a,1} > P_{b,1}$ holds, then $C_{\max}^{\pi_1} \geq C_{\max}^{\pi_2}$;*

(4) if $a \in \mathcal{G}_2$ and $b \in \mathcal{G}_2$:

- *when $\max\left\{P_{b,1} + \bar{P}_{b,2} - \bar{P}_{b,1}, P'_{b,2}\right\} \leq \max\left\{P_{a,1} + \bar{P}_{a,2} - \bar{P}_{a,1}, P'_{a,2}\right\}$ holds, then $C_{\max}^{\pi_1} \leq C_{\max}^{\pi_2}$;*
- *when $\max\left\{P_{b,1} + \bar{P}_{b,2} - \bar{P}_{b,1}, P'_{b,2}\right\} > \max\left\{P_{a,1} + \bar{P}_{a,2} - \bar{P}_{a,1}, P'_{a,2}\right\}$ holds, then $C_{\max}^{\pi_1} \geq C_{\max}^{\pi_2}$;*

where the sets $\mathcal{G}_1$ and $\mathcal{G}_2$ are defined in Definition 4.

Proof of Lemma EC.3. According to Lemma EC.1, the completion time of batch-wise schedules π_1 and π_2 are given by:

$$C_{\max}^{\pi_1} = \max\left\{\begin{array}{ll} S_{1,G,u}^{\pi_1} \triangleq \bar{C}_{G-2,1} + \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{u} P_{b,1}^{(i)} + \sum_{i=u}^{t} P_{b,2}^{(i)}, & \forall u \in [t], \\ S_{2,G,u}^{\pi_1} \triangleq \bar{C}_{G-2,1} + \sum_{i=1}^{u} P_{a,1}^{(i)} + \sum_{i=u}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{b,2}^{(i)}, & \forall u \in [r], \\ S_{3,G}^{\pi_1} \triangleq \bar{C}_{G-2,2} + \sum_{i=1}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{b,2}^{(i)} & \end{array}\right\}, \tag{EC.6}$$

$$C_{\max}^{\pi_2} = \max \left\{ \begin{array}{ll} S_{1,G,u}^{\pi_2} \triangleq \bar{C}_{G-2,1} + \sum_{i=1}^{u} P_{b,1}^{(i)} + \sum_{i=u}^{t} P_{b,2}^{(i)} + \sum_{i=1}^{r} P_{a,2}^{(i)}, & \forall u \in [t], \\ S_{2,G,u}^{\pi_2} \triangleq \bar{C}_{G-2,1} + \sum_{i=1}^{t} P_{b,1}^{(i)} + \sum_{i=1}^{u} P_{a,1}^{(i)} + \sum_{i=u}^{r} P_{a,2}^{(i)}, & \forall u \in [r], \\ S_{3,G}^{\pi_2} \triangleq \bar{C}_{G-2,2} + \sum_{i=1}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{b,2}^{(i)} & \end{array} \right\}, \tag{EC.7}$$

where

$$\bar{C}_{0,m} = 0, \quad \forall m \in \mathcal{M}.$$

We consider the following four cases.

- **Case 1:** $a \in \mathcal{G}_1$ **and** $b \in \mathcal{G}_2$**.**

  In this case, we have

$$P_{a,1}^{(k)} \le P_{a,2}^{(k)} \quad \text{and} \quad P_{b,1}^{(k)} \ge P_{b,2}^{(k)}, \quad \forall k \ge 1. \tag{EC.8}$$

  From (EC.8), we have

$$S_{1,G,u}^{\pi_1} - S_{1,G,u}^{\pi_2} = \sum_{i=1}^{r} P_{a,1}^{(i)} - \sum_{i=1}^{r} P_{a,2}^{(i)} \le 0, \quad \forall u \in [t], \tag{EC.9}$$

$$S_{2,G,u}^{\pi_1} - S_{2,G,u}^{\pi_2} = \sum_{i=1}^{t} P_{b,2}^{(i)} - \sum_{i=1}^{t} P_{b,1}^{(i)} \le 0, \quad \forall u \in [r], \tag{EC.10}$$

$$S_{3,G}^{\pi_1} = S_{3,G}^{\pi_2}. \tag{EC.11}$$

  Therefore, for every term on the right-hand side of Eq. (EC.6), there exists at least one term on the right-hand side of Eq. (EC.7) that is at least as large, which implies that $C_{\max}^{\pi_1} \le C_{\max}^{\pi_2}$.

- **Case 2:** $a \in \mathcal{G}_2$ **and** $b \in \mathcal{G}_1$

  In this case, we have

$$P_{a,1}^{(k)} \ge P_{a,2}^{(k)} \quad \text{and} \quad P_{b,1}^{(k)} \le P_{b,2}^{(k)}, \quad \forall k \ge 1, \tag{EC.12}$$

  it follows that $S_{1,G,u}^{\pi_1} \ge S_{1,G,u}^{\pi_2}$ for each $u \in [t]$, $S_{2,G,u}^{\pi_1} \ge S_{2,G,u}^{\pi_2}$ for each $u \in [r]$, and $S_{3,G}^{\pi_1} = S_{3,G}^{\pi_2}$, which implies that $C_{\max}^{\pi_1} \ge C_{\max}^{\pi_2}$.

- **Case 3:** $a \in \mathcal{G}_1$ **and** $b \in \mathcal{G}_1$

  In this case, we have

$$P_{a,1}^{(k)} \le P_{a,2}^{(k)} \quad \text{and} \quad P_{b,1}^{(k)} \le P_{b,2}^{(k)}, \quad \forall k \ge 1. \tag{EC.13}$$

  According to Lemma EC.2, Eqs. (EC.6) and (EC.7) are simplified to:

$$C_{\max}^{\pi_1} = \max \left\{ \begin{array}{l} S_{1,G}^{\pi_1} \triangleq \bar{C}_{G-2,1} + \sum_{i=1}^{r} P_{a,1}^{(i)} + P_{b,1}^{(1)} + \sum_{i=1}^{t} P_{b,2}^{(i)}, \\ S_{2,G}^{\pi_1} \triangleq \bar{C}_{G-2,1} + P_{a,1}^{(1)} + \sum_{i=1}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{b,2}^{(i)}, \\ S_{3,G}^{\pi_1} \triangleq \bar{C}_{G-2,2} + \sum_{i=1}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{b,2}^{(i)} \end{array} \right\}, \tag{EC.14}$$

$$C_{\max}^{\pi_2} = \max \left\{ \begin{array}{l} S_{1,G}^{\pi_2} \triangleq \bar{C}_{G-2,1} + P_{b,1}^{(1)} + \sum_{i=1}^{t} P_{b,2}^{(i)} + \sum_{i=1}^{r} P_{a,2}^{(i)}, \\ S_{2,G}^{\pi_2} \triangleq \bar{C}_{G-2,1} + \sum_{i=1}^{t} P_{b,1}^{(i)} + P_{a,1}^{(1)} + \sum_{i=1}^{r} P_{a,2}^{(i)}, \\ S_{3,G}^{\pi_2} \triangleq \bar{C}_{G-2,2} + \sum_{i=1}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{b,2}^{(i)} \end{array} \right\}. \tag{EC.15}$$

  Using Eq. (EC.13), we have

$$S_{1,G}^{\pi_1} \le S_{1,G}^{\pi_2} \quad \text{and} \quad S_{3,G}^{\pi_1} = S_{3,G}^{\pi_2}. \tag{EC.16}$$

When $P_{a,1} \leq P_{b,1}$ holds, we have

$$S_{2,G}^{\pi_1} \leq S_{1,G}^{\pi_2}. \tag{EC.17}$$

From Eqs. (EC.14)-(EC.17), we have $C_{\max}^{\pi_1} \leq C_{\max}^{\pi_2}$.

When $P_{a,1} > P_{b,1}$ holds, we have

$$S_{2,G}^{\pi_1} > S_{1,G}^{\pi_2} \quad \text{and} \quad S_{2,G}^{\pi_1} \geq S_{2,G}^{\pi_2}. \tag{EC.18}$$

From Eqs. (EC.14)-(EC.16), and (EC.18), we have $C_{\max}^{\pi_1} \geq C_{\max}^{\pi_2}$.

- **Case 4:** $a \in \mathcal{G}_2$ **and** $b \in \mathcal{G}_2$

  In this case, we have

$$P_{a,1}^{(k)} > P_{a,2}^{(k)} \quad \text{and} \quad P_{b,1}^{(k)} > P_{b,2}^{(k)}, \quad \forall k \geq 1. \tag{EC.19}$$

According to Lemma EC.2, the makespan expressions in Eqs. (EC.6) and (EC.7) are simplified to:

$$C_{\max}^{\pi_1} = \max \begin{Bmatrix} S_{1,G,u}^{\pi_1} \triangleq \bar{C}_{G-2,1}^{\pi_1} + \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{u} P_{b,1}^{(i)} + \sum_{i=u}^{t} P_{b,2}^{(i)},\ \forall u \in \{1,t\}, \\ S_{2,G,u}^{\pi_1} \triangleq \bar{C}_{G-2,1}^{\pi_1} + \sum_{i=1}^{u} P_{a,1}^{(i)} + \sum_{i=u}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{b,2}^{(i)},\ \forall u \in \{1,r\}, \\ S_{3,G}^{\pi_1} \triangleq \bar{C}_{G-2,2}^{\pi_1} + \sum_{i=1}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{b,2}^{(i)} \end{Bmatrix} \tag{EC.20}$$

$$C_{\max}^{\pi_2} = \max \begin{Bmatrix} S_{1,G,u}^{\pi_2} \triangleq \bar{C}_{G-2,1}^{\pi_2} + \sum_{i=1}^{u} P_{b,1}^{(i)} + \sum_{i=u}^{t} P_{b,2}^{(i)} + \sum_{i=1}^{r} P_{a,2}^{(i)},\ \forall u \in \{1,t\}, \\ S_{2,G,u}^{\pi_2} \triangleq \bar{C}_{G-2,1}^{\pi_2} + \sum_{i=1}^{t} P_{b,1}^{(i)} + \sum_{i=1}^{u} P_{a,1}^{(i)} + \sum_{i=u}^{r} P_{a,2}^{(i)},\ \forall u \in \{1,r\}, \\ S_{3,G}^{\pi_2} \triangleq \bar{C}_{G-2,2}^{\pi_2} + \sum_{i=1}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{b,2}^{(i)} \end{Bmatrix}. \tag{EC.21}$$

Using Eq. (EC.19), we have

$$S_{2,G,u}^{\pi_1} < S_{2,G,u}^{\pi_2}, \quad \forall u \in \{1,r\}, \quad \text{and} \quad S_{3,G}^{\pi_1} = S_{3,G}^{\pi_2}. \tag{EC.22}$$

When $\max\left\{P_{b,1} + \bar{P}_{b,2} - \bar{P}_{b,1}, P'_{b,2}\right\} \leq \max\left\{P_{a,1} + \bar{P}_{a,2} - \bar{P}_{a,1}, P'_{a,2}\right\}$ holds, we have

$$\max\left\{S_{1,G,v}^{\pi_1} | v \in \{1,t\}\right\} \leq \max\left\{S_{2,G,u}^{\pi_2} | u \in \{1,r\}\right\}. \tag{EC.23}$$

From Eqs. (EC.20)-(EC.23), we have $C_{\max}^{\pi_1} \leq C_{\max}^{\pi_2}$.

When $\max\left\{P_{b,1} + \bar{P}_{b,2} - \bar{P}_{b,1}, P'_{b,2}\right\} \geq \max\left\{P_{a,1} + \bar{P}_{a,2} - \bar{P}_{a,1}, P'_{a,2}\right\}$ holds, we have

$$\max\left\{S_{1,G,v}^{\pi_1} | v \in \{1,t\}\right\} > \max\left\{S_{2,G,u}^{\pi_2} | u \in \{1,r\}\right\}, \tag{EC.24}$$

$$S_{1,G,u}^{\pi_1} > S_{1,G,u}^{\pi_2}, \quad \forall u \in \{1,t\}. \tag{EC.25}$$

From Eqs. (EC.20)-(EC.22), (EC.24) and (EC.25), we have $C_{\max}^{\pi_1} \geq C_{\max}^{\pi_2}$.

■

Using the following Lemma EC.4, the conclusion of Lemma EC.3 can be generalized to any two adjacent batches in any batch-wise schedule. Let $\bar{C}_{g,m}^{\pi}$ denote the completion time of the g-th batch in a batch-wise schedule π on machine m.

LEMMA EC.4. *For the $F2/Prmu$, $CLFEs/C_{\max}$ problem with G job categories, if two batch-wise schedules π_1 and π_2 are identical from the $(g+1)$-th batch onward $(0 \le g < G)$ and satisfy $\bar{C}^{\pi_1}_{g,2} \le \bar{C}^{\pi_2}_{g,2}$, then $C^{\pi_1}_{\max} \le C^{\pi_2}_{\max}$.*

Proof of Lemma EC.4. Suppose the problem contains n jobs, and the last job in the g-th batch is the t-th job to be processed in both schedules π_1 and π_2, where $0 \le t < n$. According to the assumption $\bar{C}^{\pi_1}_{g,2} \le \bar{C}^{\pi_2}_{g,2}$, we have $C^{\pi_1}_{t,2} \le C^{\pi_2}_{t,2}$.

Since the set of the first g batches is the same in both schedules (only their order differs), the completion times of the g-th batch on the first machine for both schedules are equal, i.e., $\bar{C}^{\pi_1}_{g,1} = \bar{C}^{\pi_2}_{g,1}$, which implies that $C^{\pi_1}_{t,1} = C^{\pi_2}_{t,1}$.

Let $p_{t+1,m}$ be the actual PT of the $(t+1)$-th job processed in both schedules, where $m \in \{1,2\}$. According to Lemma EC.1, the completion times of the $(t+1)$-th job are given by

$$C^{\pi_1}_{t+1,2} = \max\left\{C^{\pi_1}_{t,1} + p_{t+1,1}, C^{\pi_1}_{t,2}\right\} + p_{t+1,2}, \tag{EC.26}$$

$$C^{\pi_2}_{t+1,2} = \max\left\{C^{\pi_2}_{t,1} + p_{t+1,1}, C^{\pi_2}_{t,2}\right\} + p_{t+1,2}. \tag{EC.27}$$

According to $C^{\pi_1}_{t,1} = C^{\pi_2}_{t,1}$ and $C^{\pi_1}_{t,2} \le C^{\pi_2}_{t,2}$, we have

$$\max\left\{C^{\pi_1}_{t,1} + p_{t+1,1}, C^{\pi_1}_{t,2}\right\} \le \max\left\{C^{\pi_2}_{t,1} + p_{t+1,1}, C^{\pi_2}_{t,2}\right\},$$

which implies that $C^{\pi_1}_{t+1,2} \le C^{\pi_2}_{t+1,2}$.

Repeating these steps up to the last job in both schedules, we have $C^{\pi_1}_{n,2} \le C^{\pi_2}_{n,2}$, which implies that $C^{\pi_1}_{\max} \le C^{\pi_2}_{\max}$.

■

Proof of Theorem 1. Denote the batch-wise schedule constructed by the BJ rule as π_{BJ}. We prove that the π_{BJ} is an optimal batch-wise schedule by contradiction.

Suppose π_{BJ} is not an optimal batch-wise schedule for the $F2$/Prmu, CLFEs/$C_{\max}$ problem. Then, it is straightforward that there exists an optimal batch-wise schedule π_0, where $C^{\pi_0}_{\max} < C^{\pi_{\mathrm{BJ}}}_{\max}$, which contains at least one pair of adjacent batches a and b satisfying one of the following conditions, according to Lemma EC.3:

- $a \in \mathcal{G}_2$ and $b \in \mathcal{G}_1$;
- $a \in \mathcal{G}_1$ and $b \in \mathcal{G}_1$, $P_{a,1} > P_{b,1}$;
- $a \in \mathcal{G}_2$ and $b \in \mathcal{G}_2$, $\max\left\{P_{b,1} + \bar{P}_{b,2} - \bar{P}_{b,1}, P'_{b,2}\right\} > \max\left\{P_{a,1} + \bar{P}_{a,2} - \bar{P}_{a,1}, P'_{a,2}\right\}$.

Let the batch b be the g-th processed batch in schedule π_0 $(1 < g \le G)$. By swapping the positions of batches a and b in schedule π_0, we obtain a new schedule π_1. According to Lemma EC.3, we have:

$$\bar{C}^{\pi_0}_{g,2} \ge \bar{C}^{\pi_1}_{g,2}.$$

Since the batches following the g-th batch are the same in both schedules π_0 and π_1, by Lemma EC.4, we have:

$$C_{\max}^{\pi_0} \geq C_{\max}^{\pi_1}.$$

Repeating this process for adjacent batches in schedule π_1 that meet the above conditions (if any), and denoting the batch-wise schedule obtained after the i-th repetition as π_i. Suppose that after T ($T \in \mathbb{Z}_+$) swaps, no adjacent batches in schedule π_T satisfy the above conditions, which implies that the schedule π_T has the same structure as π_{BJ}. Thus, for all π_i where $i = 1, \ldots, T$, we have:

$$C_{\max}^{\pi_0} \geq C_{\max}^{\pi_1} \geq \cdots \geq C_{\max}^{\pi_i} \geq \cdots \geq C_{\max}^{\pi_T} = C_{\max}^{\pi_{\text{BJ}}}.$$

Thus, we have contradiction with $C_{\max}^{\pi_0} < C_{\max}^{\pi_{\text{BJ}}}$. Therefore, the schedule π_{BJ} constructed by BJ rule is an optimal batch-wise schedule of the $F2/\text{Prmu, CLFEs}/C_{\max}$ problem. ■

EC.3.2 Proof of Theorem 2.

We first introduce some related notations and lemmas.

For any $F2/\text{Prmu, CLSFEs}/C_{\max}$ problem with G categories and n jobs ($G \geq 2, n \geq G+1$), we denote a category containing more than one job as a. We first examine a schedule π_1 of this problem that has the following structure:

$$\pi_1 = \Big(\underbrace{\overset{A_1}{j_a, \cdots, j_a}}_{r}, \underbrace{\overset{B}{j_\circ, \cdots, j_\circ}}_{s}, \underbrace{\overset{A_2}{j_a, \cdots, j_a}}_{t} \Big). \tag{EC.28}$$

The schedule π_1 is partitioned into three segments: A_1, B, and A_2. Specifically, segments A_1 and A_2 contain r and t jobs of category a (denoted as j_a, $r \geq 1, t \geq 1$), respectively. Segment B contains s jobs ($s \geq 1$) of any category (denoted as $j_\circ$), with the first and last jobs in this segment not being of category a. This schedule is denoted as $\pi_1 = (A_1, B, A_2)$.

Let the actual PT of the i-th job in segment B on machine m be $b_m^{(i)}$, for any $i \in [s]$ and $m \in \mathcal{M}$. There are two adjustments to schedule π_1 that enable the consecutive processing of jobs of category a at the beginning and end of π_1, which are $\pi_2 = (A_1, A_2, B)$ and $\pi_3 = (B, A_1, A_2)$. According to Lemma EC.1, in the $F2/\text{Prmu,CLSFEs}/C_{\max}$ problem, the completion times of schedule π_1, π_2, and π_3 are given by the following Eqs. (EC.29), (EC.30), and (EC.31), respectively:

$$C_{\max}^{\pi_1} = \max \left\{ \begin{array}{ll} S_{1,p}^{\pi_1} \triangleq \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{s} b_1^{(i)} + \sum_{i=1}^{p} P_{a,1}^{(i)} + \sum_{i=p}^{t} P_{a,2}^{(i)}, & \forall p \in [t], \\ S_{2,p}^{\pi_1} \triangleq \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{p} b_1^{(i)} + \sum_{i=p}^{s} b_2^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(i)}, & \forall p \in [s], \\ S_{3,p}^{\pi_1} \triangleq \sum_{i=1}^{p} P_{a,1}^{(i)} + \sum_{i=p}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{s} b_2^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(i)}, & \forall p \in [r] \end{array} \right\}, \tag{EC.29}$$

$$C_{\max}^{\pi_2} = \max \left\{ \begin{array}{ll} S_{1,p}^{\pi_2} \triangleq \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{p} P_{a,1}^{(r+i)} + \sum_{i=p}^{t} P_{a,2}^{(r+i)} + \sum_{i=1}^{s} b_2^{(i)}, & \forall p \in [t], \\ S_{2,p}^{\pi_2} \triangleq \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{t} P_{a,1}^{(r+i)} + \sum_{i=1}^{p} b_1^{(i)} + \sum_{i=p}^{s} b_2^{(i)}, & \forall p \in [s], \\ S_{3,p}^{\pi_2} \triangleq \sum_{i=1}^{p} P_{a,1}^{(i)} + \sum_{i=p}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(r+i)} + \sum_{i=1}^{s} b_2^{(i)}, & \forall p \in [r] \end{array} \right\}, \tag{EC.30}$$

$$C_{\max}^{\pi_3} = \max \left\{ \begin{array}{ll} S_{1,p}^{\pi_3} \triangleq \sum_{i=1}^{s} b_1^{(i)} + \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{p} P_{a,1}^{(r+i)} + \sum_{i=p}^{t} P_{a,2}^{(r+i)}, & \forall p \in [t], \\ S_{2,p}^{\pi_3} \triangleq \sum_{i=1}^{p} b_1^{(i)} + \sum_{i=p}^{s} b_2^{(i)} + \sum_{i=1}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(r+i)}, & \forall p \in [s], \\ S_{3,p}^{\pi_3} \triangleq \sum_{i=1}^{s} b_1^{(i)} + \sum_{i=1}^{p} P_{a,1}^{(i)} + \sum_{i=p}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(r+i)}, & \forall p \in [r] \end{array} \right\}. \tag{EC.31}$$

Lemma EC.5. *In the* $F2/\text{Prmu,CLSFEs}/C_{\max}$ *problem, if*

$$\sum_{i=r+1}^{r+p} P_{a,1}^{(i)} \leq \sum_{i=1}^{p} P_{a,2}^{(i)}, \quad \forall p \in [t]\,, \tag{EC.32}$$

then $C_{\max}^{\pi_2} \leq C_{\max}^{\pi_1}$.

Proof of Lemma EC.5. Due to the learning effect, we have

$$\sum_{i=p}^{t} P_{a,2}^{(r+i)} \leq \sum_{i=p}^{t} P_{a,2}^{(i)}, \quad \forall p \in [t]\,. \tag{EC.33}$$

When the inequalities (EC.32) and (EC.33) hold, we have

$$\begin{aligned}
S_{1,p}^{\pi_2} - S_{3,r}^{\pi_1} &= \left(\sum_{i=r+1}^{r+p} P_{a,1}^{(i)} + \sum_{i=r+p}^{r+t} P_{a,2}^{(i)}\right) - \left(P_{a,2}^{(r)} + \sum_{i=1}^{t} P_{a,2}^{(i)}\right) \\
&= \left(\sum_{i=r+1}^{r+p} P_{a,1}^{(i)} - \sum_{i=1}^{p} P_{a,2}^{(i)}\right) - \left(P_{a,2}^{(r)} + \sum_{i=p+1}^{t} P_{a,2}^{(i)} - \sum_{i=r+p}^{r+t} P_{a,2}^{(i)}\right) \\
&= \left(\sum_{i=r+1}^{r+p} P_{a,1}^{(i)} - \sum_{i=1}^{p} P_{a,2}^{(i)}\right) - \left(P_{a,2}^{(r)} - P_{a,2}^{(r+p)} + \sum_{i=p+1}^{t} P_{a,2}^{(i)} - \sum_{i=r+p+1}^{r+t} P_{a,2}^{(i)}\right) \\
&\leq \sum_{i=r+1}^{r+p} P_{a,1}^{(i)} - \sum_{i=1}^{p} P_{a,2}^{(i)} \leq 0, \quad \forall p \in [t]\,.
\end{aligned} \tag{EC.34}$$

From Eq. (EC.33), we have

$$\begin{aligned}
S_{2,p}^{\pi_2} &= \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{t} P_{a,1}^{(r+i)} + \sum_{i=1}^{p} b_1^{(i)} + \sum_{i=p}^{s} b_2^{(i)} \\
&\leq \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(i)} + \sum_{i=1}^{p} b_1^{(i)} + \sum_{i=p}^{s} b_2^{(i)} = S_{2,p}^{\pi_1}, \quad \forall p \in [s]\,,
\end{aligned} \tag{EC.35}$$

and

$$\begin{aligned}
S_{3,p}^{\pi_2} &= \sum_{i=1}^{p} P_{a,1}^{(i)} + \sum_{i=p}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{s} b_2^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(r+i)} \\
&\leq \sum_{i=1}^{p} P_{a,1}^{(i)} + \sum_{i=p}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{s} b_2^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(i)} = S_{3,p}^{\pi_1}, \quad \forall p \in [r]\,.
\end{aligned} \tag{EC.36}$$

Eqs. (EC.34), (EC.35), and (EC.36) show that when the inequality (EC.32) holds, we have

$$C_{\max}^{\pi_2} = \max\left\{\begin{matrix} S_{1,1}^{\pi_2}, \cdots, S_{1,t}^{\pi_2}, \\ S_{2,1}^{\pi_2}, \cdots, S_{2,s}^{\pi_2}, \\ S_{3,1}^{\pi_2}, \cdots, S_{3,r}^{\pi_2} \end{matrix}\right\} \leq \max\left\{\begin{matrix} S_{1,1}^{\pi_1}, \cdots, S_{1,t}^{\pi_1}, \\ S_{2,1}^{\pi_1}, \cdots, S_{2,s}^{\pi_1}, \\ S_{3,1}^{\pi_1}, \cdots, S_{3,r}^{\pi_1} \end{matrix}\right\} = C_{\max}^{\pi_1}. \tag{EC.37}$$

■

Lemma EC.6. *In the* $F2/\text{Prmu,CLSFEs}/C_{\max}$ *problem, if*

$$\sum_{i=p}^{r} P_{a,2}^{(i)} \leq P_{a,1}^{(1)} + \sum_{i=p+1}^{r} P_{a,1}^{(i)}, \quad \forall p \in [r]\,, \tag{EC.38}$$

then $C_{\max}^{\pi_3} \leq C_{\max}^{\pi_1}$.

Proof of Lemma EC.6. Due to the learning effect, we have

$$\sum_{i=1}^{p} P_{a,1}^{(r+i)} \le \sum_{i=1}^{p} P_{a,1}^{(i)}, \quad \forall p \in [t], \tag{EC.39}$$

$$\sum_{i=p}^{t} P_{a,2}^{(r+i)} \le \sum_{i=p}^{t} P_{a,2}^{(i)}, \quad \forall p \in [t]. \tag{EC.40}$$

From Eqs. (EC.39) and (EC.40), we have

$$\begin{aligned} S_{1,p}^{\pi_3} &= \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{p} P_{a,1}^{(r+i)} + \sum_{i=p}^{t} P_{a,2}^{(r+i)} + \sum_{i=1}^{s} b_1^{(i)} \\ &\le \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{p} P_{a,1}^{(i)} + \sum_{i=p}^{t} P_{a,2}^{(i)} + \sum_{i=1}^{s} b_1^{(i)} = S_{1,p}^{\pi_1}, \quad \forall p \in [t]. \end{aligned} \tag{EC.41}$$

When the condition (EC.38) holds for any $p \in [r]$, it follows that $\sum_{i=1}^{r} P_{a,2}^{(i)} \le \sum_{i=1}^{r} P_{a,1}^{(i)}$ when $p = 1$. Thus, we have

$$\begin{aligned} S_{2,p}^{\pi_3} &= \sum_{i=1}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(r+i)} + \sum_{i=1}^{p} b_1^{(i)} + \sum_{i=p}^{s} b_2^{(i)} \\ &\le \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(i)} + \sum_{i=1}^{p} b_1^{(i)} + \sum_{i=p}^{s} b_2^{(i)} = S_{2,p}^{\pi_1}, \quad \forall p \in [s]. \end{aligned} \tag{EC.42}$$

From Eqs. (EC.38) and (EC.40), we have

$$\begin{aligned} S_{3,p}^{\pi_3} - S_{1,1}^{\pi_1} &= \left(\sum_{i=1}^{p} P_{a,1}^{(i)} + \sum_{i=p}^{r} P_{a,2}^{(i)} + \sum_{i=r+1}^{r+t} P_{a,2}^{(i)} \right) - \left(\sum_{i=1}^{r} P_{a,1}^{(i)} + P_{a,1}^{(1)} + \sum_{i=1}^{t} P_{a,2}^{(i)} \right) \\ &= \left(\sum_{i=r+1}^{r+t} P_{a,2}^{(i)} - \sum_{i=1}^{t} P_{a,2}^{(i)} \right) - \left(P_{a,1}^{(1)} + \sum_{i=p+1}^{r} P_{a,1}^{(i)} - \sum_{i=p}^{r} P_{a,2}^{(i)} \right) \\ &\le \sum_{i=p}^{r} P_{a,2}^{(i)} - \left(P_{a,1}^{(1)} + \sum_{i=p+1}^{r} P_{a,1}^{(i)} \right) \le 0, \quad \forall p \in [r]. \end{aligned} \tag{EC.43}$$

Eqs. (EC.41), (EC.42), and (EC.43) show that when inequality (EC.38) holds, we have

$$C_{\max}^{\pi_3} = \max \left\{ \begin{matrix} S_{1,1}^{\pi_3}, \cdots, S_{1,t}^{\pi_3}, \\ S_{2,1}^{\pi_3}, \cdots, S_{2,s}^{\pi_3}, \\ S_{3,1}^{\pi_3}, \cdots, S_{3,r}^{\pi_3} \end{matrix} \right\} \le \max \left\{ \begin{matrix} S_{1,1}^{\pi_1}, \cdots, S_{1,t}^{\pi_1}, \\ S_{2,1}^{\pi_1}, \cdots, S_{2,s}^{\pi_1}, \\ S_{3,1}^{\pi_1}, \cdots, S_{3,r}^{\pi_1} \end{matrix} \right\} = C_{\max}^{\pi_1}. \tag{EC.44}$$

■

With the above analyses, we next prove the Lemma EC.7.

LEMMA EC.7. *For the $F2/Prmu,CLSFEs/C_{\max}$ problem with the schedule π_1, we have*

- *if $P_{a,1} \le P_{a,2}$, then $C_{\max}^{\pi_2} \le C_{\max}^{\pi_1}$;*
- *if $P_{a,1} > P_{a,2}$, then $C_{\max}^{\pi_3} \le C_{\max}^{\pi_1}$.*

Proof of Lemma EC.7. We consider the following two cases.

- **Case 1: $P_{a,1} \le P_{a,2}$.**

  In this case, it is straightforward that

$$P_{a,1}^{(k)} \le P_{a,2}^{(k)}, \quad \forall k \ge 1. \tag{EC.45}$$

  Due to the learning effect, we have $P_{a,1}^{(r+i)} \le P_{a,1}^{(i)}$ (where $r \ge 1$), it follows that

$$\sum_{i=1}^{p} P_{a,1}^{(r+i)} \le \sum_{i=1}^{p} P_{a,1}^{(i)} \le \sum_{i=1}^{p} P_{a,2}^{(i)}, \quad \forall p \in [t]\,. \tag{EC.46}$$

  Therefore, the condition in Lemma EC.5 is satisfied. According to Lemma EC.5, when $P_{a,1} \le P_{a,2}$, $C_{\max}^{\pi_2} \le C_{\max}^{\pi_1}$ holds.

- **Case 2: $P_{a,1} > P_{a,2}$.** In this case, it is straightforward that $P_{a,1}^{(k)} \ge P_{a,2}^{(k)}$ (where $k \ge 1$). Thus, for all $p \in [r]$, we have $P_{a,1} \ge P_{a,2} \ge P_{a,2}^{(p)}$, it follows that

$$\sum_{i=p}^{r} P_{a,2}^{(i)} \le \sum_{i=p}^{r} P_{a,1}^{(i)} \le P_{a,1}^{(1)} + \sum_{i=p+1}^{r} P_{a,1}^{(i)}, \quad \forall p \in [r]\,.$$

  Therefore, the condition in Lemma EC.6 is satisfied. According to Lemma EC.6, when $P_{a,1} > P_{a,2}$, $C_{\max}^{\pi_3} \le C_{\max}^{\pi_1}$ holds.

■

Without loss of generality, we further generalize Lemma EC.7 by extending the schedule π_1 to the following structure:

$$\pi_1' = \Big(\underbrace{\overset{C}{j_\star, \cdots, j_\star}}_{q}, \underbrace{\overset{A_1}{j_a, \cdots, j_a}}_{r}, \underbrace{\overset{B}{j_\circ, \cdots, j_\circ}}_{s}, \underbrace{\overset{A_2}{j_a, \cdots, j_a}}_{t}, \underbrace{\overset{D}{j_*, \cdots, j_*}}_{u} \Big).$$

Let $\pi_1' = (C, A_1, B, A_2, D)$, where segments C and D contain q and u ($q \ge 0, u \ge 0$) jobs of any category (denoted as $j_\star$ and j_*), respectively. If $q \ge 1$, suppose the last job in segment C is not of category a; if $u \ge 1$, suppose the first job in segment D is not of category a. Let $c_m^{(i)}$ and $d_m^{(i)}$ denote the actual PTs of i-th job in segments C and D in schedule π_1', respectively, for all $m \in \{1, 2\}$ and $i \ge 1$. In particular, we define $c_m^{(0)} = 0$ and $d_m^{(0)} = 0$.

The makespan of the schedule π_1' in the $F2$/Prmu, CLFEs/$C_{\max}$ problem is given by

$$C_{\max}^{\pi_1'} = \max\left\{ C_{\max}^{\pi_1} + \textstyle\sum_{i=0}^{q} c_2^{(i)} + \sum_{i=0}^{u} d_2^{(i)}, S_4^{\pi_1'}, S_5^{\pi_1'} \right\}, \tag{EC.47}$$

where

$$S_4^{\pi_1'} \triangleq \begin{cases} \max\limits_{p \in [u]} \Big\{ \sum\limits_{i=0}^{r} c_1^{(i)} + \sum\limits_{i=1}^{r} P_{a,1}^{(i)} + \sum\limits_{i=1}^{s} b_1^{(i)} \\ \qquad + \sum\limits_{i=1}^{t} P_{a,1}^{(i)} + \sum\limits_{i=1}^{p} d_1^{(i)} + \sum\limits_{i=p}^{u} d_2^{(i)} \Big\}, & \text{if } u \ge 1 \\ 0, & \text{otherwise,} \end{cases} \tag{EC.48}$$

$$S_5^{\pi_1'} \triangleq \begin{cases} \max\limits_{p\in[q]}\left\{ \sum\limits_{i=1}^{p} c_1^{(i)} + \sum\limits_{i=p}^{q} c_2^{(i)} + \sum\limits_{i=1}^{r} P_{a,2}^{(i)} \right. \\ \qquad\qquad \left. + \sum\limits_{i=1}^{s} b_2^{(i)} + \sum\limits_{i=1}^{t} P_{a,2}^{(i)} + \sum\limits_{i=0}^{u} d_2^{(i)} \right\}, & \text{if } q \ge 1 \\ 0, & \text{otherwise.} \end{cases} \tag{EC.49}$$

By moving segments A_1 and A_2 in the schedule π_1', we obtain two schedules $\pi_2' = (C, A_1, A_2, B, D)$ and $\pi_3' = (C, B, A_1, A_2, D)$, where segments A_1 and A_2 are processed consecutively. Then, the following conclusion holds.

Lemma EC.8. *For the $F2/\mathrm{Prmu,CLSFEs}/C_{\max}$ problem with a schedule $\pi_1' = (C, A_1, B, A_2, D)$ where $q \ge 0, r \ge 1, s \ge 1, t \ge 1, u \ge 0$:*

- *if $P_{a,1} \le P_{a,2}$, then $C_{\max}^{\pi_2'} \le C_{\max}^{\pi_1'}$;*
- *if $P_{a,1} > P_{a,2}$, then $C_{\max}^{\pi_3'} \le C_{\max}^{\pi_1'}$.*

Proof of Lemma EC.8. We discuss the following two cases.

- **Case 1: $P_{a,1} \le P_{a,2}$.**

  The completion times of π_2' is given by Eq. (EC.50).

$$C_{\max}^{\pi_2'} = \max\left\{ C_{\max}^{\pi_2} + \textstyle\sum_{i=0}^{q} c_1^{(i)} + \sum_{i=0}^{u} d_2^{(i)\prime}, S_4^{\pi_2'}, S_5^{\pi_2'} \right\}, \tag{EC.50}$$

where

$$S_4^{\pi_2'} \triangleq \begin{cases} \max\limits_{p\in[u]}\left\{ \sum\limits_{i=0}^{q} c_1^{(i)} + \sum\limits_{i=1}^{r} P_{a,1}^{(i)} + \sum\limits_{i=1}^{t} P_{a,1}^{(r+i)} \right. \\ \qquad\qquad \left. + \sum\limits_{i=1}^{s} b_1^{(i)} + \sum\limits_{i=1}^{p} d_1'^{(i)} + \sum\limits_{i=p}^{u} d_2'^{(i)} \right\}, & \text{if } u \ge 1 \\ 0, & \text{otherwise,} \end{cases} \tag{EC.51}$$

$$S_5^{\pi_2'} \triangleq \begin{cases} \max\limits_{p\in[q]}\left\{ \sum\limits_{i=1}^{p} c_1^{(i)} + \sum\limits_{i=p}^{q} c_2^{(i)} + \sum\limits_{i=1}^{r} P_{a,2}^{(i)} \right. \\ \qquad\qquad \left. + \sum\limits_{i=1}^{t} P_{a,2}^{(r+i)} + \sum\limits_{i=1}^{s} b_2^{(i)} + \sum\limits_{i=0}^{u} d_2'^{(i)} \right\}, & \text{if } q \ge 1 \\ 0, & \text{otherwise,} \end{cases} \tag{EC.52}$$

where the $d_m^{(i)\prime}$ denote the actual PT of i-th job in segment D after the movement of A_1, for all $m \in \{1, 2\}$ and $i \in [u]$. We have

$$d_m^{(i)\prime} \le d_m^{(i)}, \quad \forall m \in \{1, 2\}, i \in [u], \tag{EC.53}$$

since if the category of the first job in segment D is the same as the last one in segment B, the actual PT of jobs in segment D may decrease due to the learning effect.

According to the learning effect, we have

$$\sum_{i=1}^{t} P_{a,1}^{(r+i)} \le \sum_{i=1}^{t} P_{a,1}^{(i)} \quad \text{and} \quad \sum_{i=1}^{t} P_{a,2}^{(r+i)} \le \sum_{i=1}^{t} P_{a,2}^{(i)},$$

it follows that

$$S_4^{\pi_2'} \leq S_4^{\pi_1'} \quad \text{and} \quad S_5^{\pi_2'} \leq S_5^{\pi_1'}. \tag{EC.54}$$

When $P_{a,1} \leq P_{a,2}$ holds, applying Lemma EC.7, we have

$$C_{\max}^{\pi_2} \leq C_{\max}^{\pi_1}. \tag{EC.55}$$

In summary, using Eqs. (EC.47), (EC.50), (EC.53), (EC.54), and (EC.55), we have

$$C_{\max}^{\pi_2'} \leq C_{\max}^{\pi_1'}. \tag{EC.56}$$

- **Case 2:** $P_{a,1} > P_{a,2}$**.**
  In this case, we can prove $C_{\max}^{\pi_3'} \leq C_{\max}^{\pi_1'}$ in a similar way.

■

Proof of Theorem 2. Denote the set of optimal schedules for the $F2/\text{Prmu, CLSFEs}/C_{\max}$ problem as Π. We prove the theorem by contradiction. Suppose for any batch-wise schedule π, we have $\pi \notin \Pi$.

According to Lemma EC.8, any non-batch-wise schedule π' for the $F2/\text{Prmu, CLSFEs}/C_{\max}$ problem can be adjusted into a batch-wise schedule π through a finite number of interchanges with $C_{\max}^{\pi} \leq C_{\max}^{\pi'}$. The steps are as follows:

(1) Identify the first category from the beginning of the schedule whose jobs are not all processed consecutively. Let this category be a.
(2) Let A_1 and A_2 denote the first and last consecutive segments of jobs from category a, respectively.
(3) According to Lemma EC.8:
   - if $P_{a,1} \leq P_{a,2}$, move segment A_2 to follow segment A_1;
   - if $P_{a,1} > P_{a,2}$, move segment A_1 to precede segment A_2.
(4) Repeat steps (1)-(3) until all jobs of each category are processed consecutively, resulting in a batch-wise schedule π.

Thus, we can transform any optimal schedule π_1 ($\pi_1 \in \Pi$) into a batch-wise schedule π_2 such that $C_{\max}^{\pi_2} \leq C_{\max}^{\pi_1}$. Consequently, we find a batch-wise schedule $\pi_2 \in \Pi$. We hence have a contradiction with $\pi_2 \notin \Pi$. Therefore, Theorem 2 is true. ■

EC.3.3 Proof of Lemma 3.

To prove Lemma 3, we first introduce the following result.

LEMMA EC.9. *For the $F2/Prmu, CLE/C_{\max}$ problem with n jobs, if two schedules π_1 and π_2 are identical from the $(k+1)$-th position onward $(0 \leq k < n)$ and satisfy $C_{k,m}^{\pi_1} \geq C_{k,m}^{\pi_2}$ for all $m \in \{1,2\}$, then $C_{\max}^{\pi_1} \geq C_{\max}^{\pi_2}$.*

Proof of Lemma EC.9. Let $p^{\pi}_{k,m}$ denote the actual PT of the k-th job on the m-th machine in schedule π. According to Lemma EC.1, the completion times of the $(k+1)$-th job in schedules π_1 and π_2 are given by Eqs. (EC.57) and (EC.58), respectively.

$$C^{\pi_1}_{k+1,2} = \max\left\{C^{\pi_1}_{k,1} + p^{\pi_1}_{k+1,1}, C^{\pi_1}_{k,2}\right\} + p^{\pi_1}_{k+1,2}, \tag{EC.57}$$

$$C^{\pi_2}_{k+1,2} = \max\left\{C^{\pi_2}_{k,1} + p^{\pi_2}_{k+1,1}, C^{\pi_2}_{k,2}\right\} + p^{\pi_2}_{k+1,2}. \tag{EC.58}$$

Since there is no forgetting effect in this problem, the actual PTs of the $(k+1)$-th job and all subsequent jobs are identical in both schedules, that is,

$$p^{\pi_1}_{t,m} = p^{\pi_2}_{t,m}, \quad \forall t \in \{k+1, \cdots, n\}, m \in \{1, 2\}, \tag{EC.59}$$

and

$$C^{\pi_1}_{k,m} \geq C^{\pi_2}_{k,m}, \quad \forall m \in \{1, 2\}. \tag{EC.60}$$

From Eqs. (EC.57)-(EC.60), we have

$$C^{\pi_1}_{k+1,2} \geq C^{\pi_2}_{k+1,2}. \tag{EC.61}$$

If $k+1 = n$, then $C^{\pi_1}_{\max} \geq C^{\pi_2}_{\max}$ is proved. If $k+1 \leq n$, repeating this process from position $k+2$ until the last job in the schedule results in $C^{\pi_1}_{\max} \geq C^{\pi_2}_{\max}$. ■

Proof of Lemma 3. Let $p^{\pi}_{k,m}$ denote the actual PT of the k-th job on machine m in schedule π. For the $F2/\mathrm{Prmu}, \mathrm{CLE}/C_{\max}$ problem with n jobs and non-empty job subsets $\mathcal{J}_1$ and $\mathcal{J}_2$, let Π denote the set of optimal schedules for the problem, and Π' denote the set of schedules satisfying the given structure. We prove $\Pi \cap \Pi' \neq \emptyset$ by contradiction. Assume to the contrary that $\Pi \cap \Pi' = \emptyset$, so that for any schedule $\pi_0 \in \Pi$, there exist adjacent jobs a and b such that $a \in \mathcal{J}_2$ and $b \in \mathcal{J}_1$. This implies that the actual PTs of jobs a and b in schedule π_0 satisfy

$$p^{\pi_0}_{k,1} > p^{\pi_0}_{k,2} \quad \text{and} \quad p^{\pi_0}_{k+1,1} \leq p^{\pi_0}_{k+1,2}, \tag{EC.62}$$

where we assume that job a is at position k and job b is at position $k+1$ in the schedule π_0 $(1 \leq k < n)$. By swapping jobs a and b, we obtain a schedule π_1. According to Lemma EC.1, the completion times of the $(k+1)$-th job in schedules π_0 and π_1 are given by Eqs. (EC.63) and (EC.64), respectively.

$$C^{\pi_0}_{k+1,2} = \max\left\{\begin{array}{l} A_1 \triangleq C^{\pi_0}_{k-1,1} + p^{\pi_0}_{k,1} + p^{\pi_0}_{k+1,1} + p^{\pi_0}_{k+1,2}, \\ A_2 \triangleq C^{\pi_0}_{k-1,1} + p^{\pi_0}_{k,1} + p^{\pi_0}_{k,2} + p^{\pi_0}_{k+1,2}, \\ A_3 \triangleq C^{\pi_0}_{k-1,2} + p^{\pi_0}_{k,2} + p^{\pi_0}_{k+1,2} \end{array}\right\}, \tag{EC.63}$$

$$C^{\pi_1}_{k+1,2} = \max\left\{\begin{array}{l} B_1 \triangleq C^{\pi_1}_{k-1,1} + p^{\pi_1}_{k+1,1} + p^{\pi_1}_{k,1} + p^{\pi_1}_{k,2}, \\ B_2 \triangleq C^{\pi_1}_{k-1,1} + p^{\pi_1}_{k+1,1} + p^{\pi_1}_{k+1,2} + p^{\pi_1}_{k,2}, \\ B_3 \triangleq C^{\pi_1}_{k-1,2} + p^{\pi_1}_{k+1,2} + p^{\pi_1}_{k,2} \end{array}\right\}. \tag{EC.64}$$

where

$$C_{k-1,m}^{\pi_0} = C_{k-1,m}^{\pi_1}, \quad \forall m \in \mathcal{M}. \tag{EC.65}$$

Since this problem considers only the learning effect (no forgetting), the actual PTs of a job depend on its position in the sequence, not on the adjacent jobs. Thus, when swapped, the PTs of jobs a and b are preserved:

$$p_{k,m}^{\pi_1} = p_{k+1,m}^{\pi_0}, \quad \forall m \in \mathcal{M},$$

$$p_{k+1,m}^{\pi_1} = p_{k,m}^{\pi_0}, \quad \forall m \in \mathcal{M}, \tag{EC.66}$$

$$C_{k+1,1}^{\pi_0} = C_{k+1,1}^{\pi_1}. \tag{EC.67}$$

According to Eq. (EC.62), (EC.65)-(EC.66), we have $A_1 > B_2$, $A_2 \geq B_1$, and $A_3 = B_3$ in Eqs. (EC.63) and (EC.64), which implies that

$$C_{k+1,2}^{\pi_0} \geq C_{k+1,2}^{\pi_1}. \tag{EC.68}$$

According to Lemma EC.9 and Eqs. (EC.67)- (EC.68), we have

$$C_{\max}^{\pi_0} \geq C_{\max}^{\pi_1}.$$

Repeating the above steps until T adjacent job swaps have been made, such that all jobs in set $\mathcal{J}_1$ are processed before all jobs in set $\mathcal{J}_2$. Denote the resulting schedule as $\pi_T \in \Pi'$. Similarly, we have

$$C_{\max}^{\pi_0} \geq C_{\max}^{\pi_1} \geq \cdots \geq C_{\max}^{\pi_T}.$$

Since $\pi_0 \in \Pi$, it follows that $\pi_T \in \Pi$. Thus, we have contradiction with $\Pi \cap \Pi' = \emptyset$. Therefore, the Lemma 3 is true. ■

EC.3.4 Proof of Proposition 2.

Let Π denote the set of optimal schedules for jobs in subset $\mathcal{J}_1$, and Π_B denote the set of batch-wise schedules for these jobs. To prove the BJ rule is the optimal scheduling rule for jobs in subset $\mathcal{J}_1$ of the $F2/\text{Prmu, CLFEs}/C_{\max}$ problem, it suffices to show that there always exists an optimal schedule that is a batch-wise schedule for these jobs, i.e. $\Pi \cap \Pi_B \neq \emptyset$.

We prove $\Pi \cap \Pi_B \neq \emptyset$ by contradiction. Assume to the contrary that $\Pi \cap \Pi_B = \emptyset$, so that for any schedule $\pi_0 \in \Pi$, there exists at least one category whose jobs are not processed consecutively. Denote the first category not processed consecutively in the schedule π_0 as category a. Let A_1 and A_2 denote the first and last segments of jobs from category a, with r and t jobs, respectively ($r, t \geq 1$). Thus, the schedule π_0 has the following structure:

$$\pi_0 : \Big(\underbrace{\cdots, \underbrace{j_a, \overset{A_1}{\cdots}, j_a}_{r}, \underbrace{j_\circ, \overset{B}{\cdots}, j_\circ}_{s}, \underbrace{j_a, \overset{A_2}{\cdots}, j_a}_{t}, \cdots}_{|\mathcal{J}_1|}\Big),$$

where segment B contains s jobs ($s \geq 1$). Segment B may contain jobs of multiple categories, but its first and last jobs are not of category a. Let the first job in segment A_1 be the (k_1+1)-th job processed, and the last job in segment A_2 be the k_2-th job processed ($0 \leq k_1 < k_2 \leq |\mathcal{J}_1|$). The actual PT of the i-th job in segment B is denoted as $b_m^{(i)}$. By swapping segments B and A_2 in the schedule π_0, we obtain a schedule π_1, which has the following structure:

$$\pi_1 : \Big(\underbrace{\cdots, \overbrace{\underbrace{j_a, \cdots, j_a}_{r}}^{A_1}, \overbrace{\underbrace{j_a, \cdots, j_a}_{t}}^{A_2}, \overbrace{\underbrace{j_\circ, \cdots, j_\circ}_{s}}^{B}, \cdots}_{|\mathcal{J}_1|} \Big).$$

According to Lemma EC.1, the completion times of the k_2-th job in schedules π_0 and π_1 are given by:

$$C_{k_2,2}^{\pi_0} = \max \left\{ \begin{array}{l} S_{1,p}^{\pi_0} \triangleq C_{k_1,1} + \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{s} b_1^{(i)} + \sum_{i=1}^{p} P_{a,1}^{(v+i)} + \sum_{i=p}^{t} P_{a,2}^{(v+i)}, \forall p \in [t], \\ S_{2,p}^{\pi_0} \triangleq C_{k_1,1} + \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{p} b_1^{(i)} + \sum_{i=p}^{s} b_2^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(v+i)}, \quad \forall p \in [s], \\ S_{3,p}^{\pi_0} \triangleq C_{k_1,1} + \sum_{i=1}^{p} P_{a,1}^{(i)} + \sum_{i=p}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{s} b_2^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(v+i)}, \quad \forall p \in [r], \\ S_4^{\pi_0} \triangleq C_{k_1,2} + \sum_{i=1}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{s} b_2^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(v+i)} \end{array} \right\}, \tag{EC.69}$$

and

$$C_{k_2,2}^{\pi_1} = \max \left\{ \begin{array}{l} S_{1,p}^{\pi_1} \triangleq C_{k_1,1} + \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{p} P_{a,1}^{(r+i)} + \sum_{i=p}^{t} P_{a,2}^{(r+i)} + \sum_{i=1}^{s} b_2^{(i)}, \forall p \in [t], \\ S_{2,p}^{\pi_1} \triangleq C_{k_1,1} + \sum_{i=1}^{r} P_{a,1}^{(i)} + \sum_{i=1}^{t} P_{a,1}^{(r+i)} + \sum_{i=1}^{p} b_1^{(i)} + \sum_{i=p}^{s} b_2^{(i)}, \quad \forall p \in [s], \\ S_{3,p}^{\pi_1} \triangleq C_{k_1,1} + \sum_{i=1}^{p} P_{a,1}^{(i)} + \sum_{i=p}^{r} P_{a,2}^{(i)} + \sum_{i=1}^{t} P_{a,2}^{(r+i)} + \sum_{i=1}^{s} b_2^{(i)}, \quad \forall p \in [r], \\ S_4^{\pi_1} \triangleq C_{k_1,2} + \sum_{i=1}^{r+t} P_{a,2}^{(i)} + \sum_{i=1}^{s} b_2^{(i)} \end{array} \right\}, \tag{EC.70}$$

where $v = \max\{r - s\delta, 0\} \leq r$, according to Eq. (2). It can be shown that

$$\left. \begin{array}{rl} S_{3,1}^{\pi_0} \geq S_{1,p}^{\pi_0}, & \forall p \in [t] \\ S_{2,p}^{\pi_0} \geq S_{2,p}^{\pi_1}, & \forall p \in [s] \\ S_{3,p}^{\pi_0} \geq S_{3,p}^{\pi_1}, & \forall p \in [r] \\ S_4^{\pi_0} \geq S_4^{\pi_1} & \end{array} \right\} \Rightarrow C_{k_2,2}^{\pi_0} \geq C_{k_2,2}^{\pi_1}.$$

Since the category a is the first category that is not processed consecutively, moving segment A_2 will not increase the PTs of any job after segment A_1, that is,

$$C_{k_2,1}^{\pi_0} \geq C_{k_2,1}^{\pi_1}.$$

According to Lemma EC.9, we have

$$C_{|\mathcal{J}_1|,2}^{\pi_0} \geq C_{|\mathcal{J}_1|,2}^{\pi_1}.$$

Repeat the above process. After T ($T \in \mathbb{Z}_+$) iterations, all jobs of the same category are processed consecutively in the resulting schedule π_T, i.e., $\pi_T \in \Pi_B$. Thus, we have

$$C_{|\mathcal{J}_1|,2}^{\pi_0} \geq C_{|\mathcal{J}_1|,2}^{\pi_1} \geq \cdots \geq C_{|\mathcal{J}_1|,2}^{\pi_T}.$$

Since schedule π_0 is an optimal schedule for the problem, the batch-wise schedule π_T is also an optimal schedule. This implies that

$$\pi_T \in \Pi \cap \Pi_B \neq \emptyset,$$

which contradicts our initial assumption $\Pi \cap \Pi_B = \emptyset$. Therefore, the Proposition 2 is true. ■

EC.3.5 Proof of Proposition 3.

To construct an optimal schedule for jobs in subset $\mathcal{J}_2$ of the $F2/\text{Prmu, CLE}/C_{\max}$ problem, we first index the jobs within each batch $c \in C$ from 1 to n_c. We then define their equivalent PTs on each machine $m \in \mathcal{M}$ as $P_{c,m}^{(1)}, \ldots, P_{c,m}^{(n_c)}$.

Note that the schedule π_0, constructed according to Proposition 3, is characterized by two properties:

(i) All jobs are sequenced in descending order of their equivalent PTs.

(ii) Within each category, all jobs are sequenced in ascending order of their indices.

Let Π denote the set of optimal schedules for jobs in subset $\mathcal{J}_2$. We prove $\pi_0 \in \Pi$ by contradiction. Assume to the contrary that $\pi_0 \notin \Pi$, which implies that any optimal schedule $\pi \in \Pi$ must violate at least one of the properties (i)-(ii) that characterize schedule π_0. For an arbitrary optimal schedule $\pi_1 \in \Pi$, we consider the following three cases.

- **Case 1: Schedule π_1 satisfies property (i) but violates property (ii).**

  In this case, schedule π_1 does not sequence the jobs of at least one category in ascending order of their indices. We observe that for any given category, swapping the processing order of its jobs does not alter the actual PT of any job in the schedule, nor does it affect the final makespan. Therefore, we can transform schedule π_1 into schedule π_0 by re-sorting the jobs within each category into the correct ascending order of their indices. This transformation consists of a finite number of swaps and does not change the makespan of the schedule. This implies that $C_{\max}^{\pi_1} = C_{\max}^{\pi_0}$. Since π_1 is assumed to be an optimal schedule (i.e., $\pi_1 \in \Pi$), we have $\pi_0 \in \Pi$. This contradicts the initial assumption that $\pi_0 \notin \Pi$. Therefore, the schedule π_0 is optimal.

- **Case 2: Schedule π_1 satisfies property (ii) but violates property (i).**

  Let $p_{k,m}^{\pi}$ denote the actual PT of the k-th job on machine m in a schedule π. Violating property (i) implies that there exists at least one pair of adjacent jobs a and b from different categories (where $d_{a,c} + d_{b,c} \leq 1$ for all $c \in C$) that satisfies:

$$p_{k,2}^{\pi_1} < p_{k+1,2}^{\pi_1}, \tag{EC.71}$$

  where job a is at position k and job b is at position $k+1$ in the schedule π_1 ($1 \leq k < n$). Since both jobs $a, b \in \mathcal{J}_2$, we have

$$p_{e,1}^{\pi} > p_{e,2}^{\pi}, \quad \forall e \in \{k, k+1\}, \pi \in \{\pi_1, \pi_2\}.$$

  We construct a new schedule π_2 by swapping the positions of jobs a and b. The completion times for all preceding jobs remain unchanged:

$$C_{k-1,m}^{\pi_1} = C_{k-1,m}^{\pi_2}, \quad \forall m \in \mathcal{M}.$$

Since this problem considers only the learning effect (no forgetting), the actual PTs of a job depend on its position in the sequence, not on the adjacent jobs. Thus, when swapped, the PTs of jobs a and b are preserved:

$$p^{\pi_1}_{k,m} = p^{\pi_2}_{k+1,m}, \quad \forall m \in \mathcal{M},$$

$$p^{\pi_1}_{k+1,m} = p^{\pi_2}_{k,m}, \quad \forall m \in \mathcal{M}.$$

From Lemma EC.1 and Eqs. (EC.71)–(EC.72), the completion times at position $k+1$ in schedules π_1 and π_2 satisfy:

$$C^{\pi_2}_{k+1,1} = C^{\pi_1}_{k+1,1}, \tag{EC.72}$$

$$C^{\pi_2}_{k+1,2} = \max\left\{\begin{matrix} C^{\pi_2}_{k-1,1} + p^{\pi_2}_{k,1} + p^{\pi_2}_{k+1,1} + p^{\pi_2}_{k+1,2}, \\ C^{\pi_2}_{k-1,1} + p^{\pi_2}_{k,1} + p^{\pi_2}_{k,2} + p^{\pi_2}_{k+1,2}, \\ C^{\pi_2}_{k-1,2} + p^{\pi_2}_{k,2} + p^{\pi_2}_{k+1,2} \end{matrix}\right\} \leq \max\left\{\begin{matrix} C^{\pi_1}_{k-1,1} + p^{\pi_1}_{k,1} + p^{\pi_1}_{k+1,1} + p^{\pi_1}_{k+1,2}, \\ C^{\pi_1}_{k-1,1} + p^{\pi_1}_{k,1} + p^{\pi_1}_{k,2} + p^{\pi_1}_{k+1,2}, \\ C^{\pi_1}_{k-1,2} + p^{\pi_1}_{k,2} + p^{\pi_1}_{k+1,2} \end{matrix}\right\} = C^{\pi_1}_{k+1,2}. \tag{EC.73}$$

From Lemma EC.9, we have:

$$C^{\pi_2}_{\max} \leq C^{\pi_1}_{\max}. \tag{EC.74}$$

We can iteratively apply this swapping procedure to any adjacent pair of jobs that violates property (i). This process will terminate in a finite T ($T \in \mathbb{Z}_+$) steps. The final schedule, π_{T+1}, will satisfy both properties (i) and (ii), meaning $\pi_{T+1} = \pi_0$. Since each swap does not increase the makespan, we have

$$C^{\pi_1}_{\max} \geq C^{\pi_2}_{\max} \geq \cdots \geq C^{\pi_{T+1}}_{\max} = C^{\pi_0}_{\max}, \tag{EC.75}$$

which implies that π_0 is also an optimal schedule, i.e., $\pi_0 \in \Pi$. This contradicts our initial assumption that $\pi_0 \notin \Pi$. Therefore, the schedule π_0 is optimal.

- **Case 3: Schedule π_1 violates both properties (i) and (ii).**

  In this case, we can transform the schedule π_1 into the schedule π_0 in two phases, neither of which increases the makespan. First, we apply the re-sorting procedure from Case 1 to the schedule π_1. This corrects any violations of property (ii) and results in an intermediate schedule π', such that $C^{\pi'}_{\max} \leq C^{\pi_1}_{\max}$. Next, the schedule π' satisfies property (ii) but still violates property (i). We then apply the iterative swapping procedure in Case 2 to the schedule π', which corrects any violations of property (i) and transforms π' into π_0, such that $C^{\pi_0}_{\max} \leq C^{\pi'}_{\max}$. Therefore, we have $C^{\pi_0}_{\max} \leq C^{\pi'}_{\max} \leq C^{\pi_1}_{\max}$. Since the schedule π_1 is an optimal schedule by assumption, we have $\pi_0 \in \Pi$. This contradicts our initial assumption that $\pi_0 \notin \Pi$. Therefore, the schedule π_0 is optimal.

In summary, we show that Proposition 3 provides an optimal schedule for jobs in subset $\mathcal{J}_2$ of the $F2/\text{Prmu, CLE}/C_{\max}$ problem. ■

EC.3.6 Proof of Theorem 3.

According to Lemma 3, to prove that the PJ rule is the optimal scheduling rule for the $F2/\text{Prmu, CLE}/C_{\max}$ problem, it suffices to show that concatenating the optimal schedules of jobs in subsets $\mathcal{J}_1$ and $\mathcal{J}_2$ yields an optimal schedule for the original problem.

Let schedule π be an optimal schedule for the $F2/\text{Prmu, CLE}/C_{\max}$ problem that satisfies the structural property of Lemma 3. By reordering the first $|\mathcal{J}_1|$ jobs in schedule π using the BJ rule, we obtain a schedule π_1, and consequently,

$$C^{\pi}_{|\mathcal{J}_1|,1} = C^{\pi_1}_{|\mathcal{J}_1|,1},$$

$$C^{\pi}_{|\mathcal{J}_1|,2} \geq C^{\pi_1}_{|\mathcal{J}_1|,2},$$

according to Lemma EC.9, we have

$$C^{\pi}_{\max} \geq C^{\pi_1}_{\max}.$$

Thus, the schedule π_1 remains optimal. Similarly, reordering the jobs in $\mathcal{J}_2$ in π_1 using Proposition 3 yields another optimal schedule π_2. Therefore, there exists an optimal schedule (i.e., schedule π_2) for the $F2/\text{Prmu, CLE}/C_{\max}$ problem, which concatenates the optimal schedules of jobs in subsets $\mathcal{J}_1$ and $\mathcal{J}_2$. ■

EC.3.7 Proof of Theorem 4.

For the general $Fm/\text{Prmu, CLFEs}/C_{\max}$ problem, let π_{BJ} denote the schedule constructed by the BJ rule, and π_{PJ} denote the optimal schedule for the corresponding $F2/\text{Prmu, CLE}/C_{\max}$ problem (after ignoring the forgetting effect) constructed by the PJ rule. We verify these sufficient conditions separately.

(1) When $|\mathcal{G}_2| \leq 1$, according to Lemma 3 and Proposition 2, both schedules π_{BJ} and π_{PJ} are batch-wise. They both process the batches in subset $\mathcal{G}_1$ in optimal order first, and then process the batch in subset $\mathcal{G}_2$ (if any). It follows that the two schedules are identical. Therefore, the lower and upper bounds of the optimal solution coincide, i.e.,

$$C^{\pi_{\text{BJ}}}_{\max} = C^{*}_{\max} = C^{\pi_{\text{PJ}}}_{\max,\,\delta=0}.$$

Thus, in this case, the BJ rule is an optimal scheduling rule.

(2) Let π^* be the optimal schedule for the problem. When the condition (2) holds, we prove the BJ rule is an optimal scheduling rule by contradiction. Assume to the contrary that

$$C^{\pi^*}_{\max} < C^{\pi_{\text{BJ}}}_{\max}. \tag{EC.76}$$

The optimal schedule π^* may not be a batch-wise schedule, which implies that

$$C^{\pi^*}_{n,1} \geq C^{\pi_{\text{BJ}}}_{n,1}. \tag{EC.77}$$

From condition

$$C^{\pi_{\text{BJ}}}_{k,1} \geq C^{\pi_{\text{BJ}}}_{k-1,2}, \quad \forall k \in [n] \setminus [\mathcal{J}_1],$$

we have

$$\bar{P}_{c,1} + P'_{c,2} \geq \bar{P}_{c,2} + P_{c,1}, \quad \forall c \in \mathcal{G}_2,$$

and it follows that

$$\max\left\{P'_{c,2}, \bar{P}_{c,2} + P_{c,1} - \bar{P}_{c,1}\right\} = P'_{c,2}, \quad \forall c \in \mathcal{G}_2.$$

This means that the BJ rule sorts all batches in batch subset $\mathcal{G}_2$ according to $P'_{c,2}$ under this condition. Let $Q_n^{\pi_{\mathrm{BJ}}}$ and $Q_n^{\pi^*}$ denote the PTs of the n-th job on the second machine in schedules π_{BJ} and π^*, respectively. The actual PT of the last job in the schedule π_{BJ} is the minimum PT of all jobs in job subset $\mathcal{J}_2$ on the second machine, i.e.,

$$Q_n^{\pi^*} \geq Q_n^{\pi_{\mathrm{BJ}}} = \min\left\{P'_{c,2} \mid c \in \mathcal{G}_2\right\}. \tag{EC.78}$$

From Eqs. (EC.77) and (EC.78), we have

$$C_{\max}^{\pi^*} \geq C_{n,1}^{\pi^*} + Q_n^{\pi^*} \geq C_{n,1}^{\pi_{\mathrm{BJ}}} + Q_n^{\pi_{\mathrm{BJ}}} = C_{\max}^{\pi_{\mathrm{BJ}}}, \tag{EC.79}$$

This contradicts our initial assumption in Eq. (EC.76). Therefore, when the condition (2) is satisfied, the BJ rule is globally optimal for the problem.

■

References

Reza Hejazi, S, S Saghafian. 2005. Flowshop-scheduling problems with makespan criterion: a review. *International Journal of Production Research,* 43 (14), 2895-2929.